

Interim Report: Portfolio Optimization — Data Exploration & Analysis

Project: Time Series Forecasting for Portfolio Management Optimization

Client: Guide Me in Finance (GMF) Investments

Date: July 5, 2026

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Data loaded successfully.

1. Data Extraction and Cleaning

1.1 Data Source and Scope

Historical financial data was extracted from the **YFinance API** for three assets selected to represent different risk profiles:

Asset	Ticker	Description	Risk Profile
Tesla	TSLA	High-growth automobile manufacturer	High risk, high potential return
Vanguard Total Bond Market ETF	BND	Tracks U.S. investment-grade bonds	Low risk, stability and income
S&P 500 ETF	SPY	Tracks the S&P 500 Index	Moderate risk, broad market exposure

Coverage period: January 1, 2015 to June 30, 2026 (~11.5 years)

DATASET OVERVIEW

Asset	Rows	Cols	Start Date	End Date	Missing
TSLA	2888	6	2015-01-02	2026-06-29	0
BND	2888	6	2015-01-02	2026-06-29	0
SPY	2888	6	2015-01-02	2026-06-29	0

Total trading days: 2888

Fields per asset: Open, High, Low, Close, Adj Close, Volume

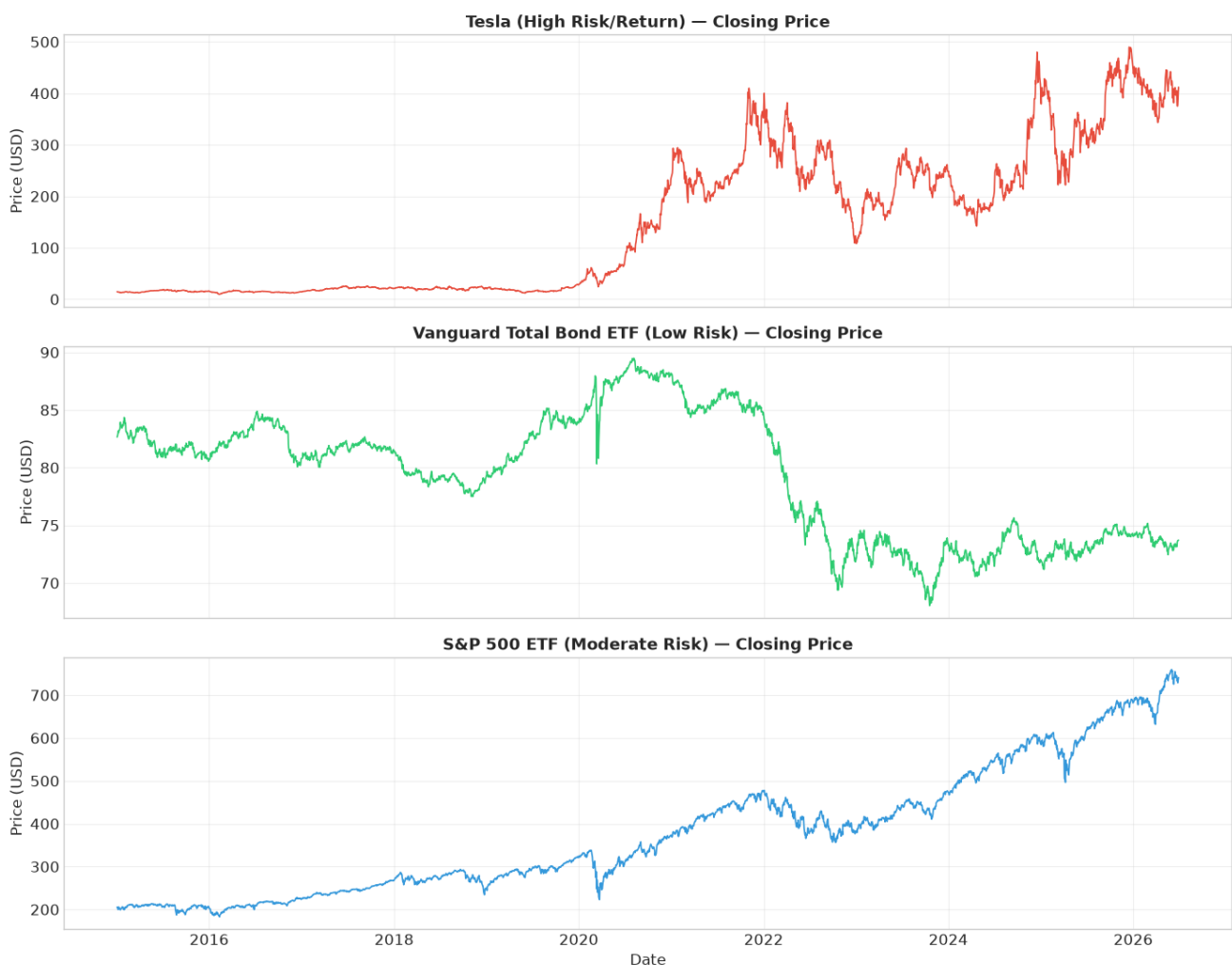
1.2 Cleaning Steps Performed

Step	Action	Result
Missing value check	Scanned all columns for NaN/null values	No missing values found in any of the three datasets

Step	Action	Result
Data type validation	Verified all price columns are float64, index is DatetimeIndex	All types correct; indices are monotonically increasing
Temporal alignment	Checked that all three assets share the same trading dates	2,888 common trading days
Derived features	Computed daily percentage returns: <code>pct_change()</code>	Created <code>daily_returns</code> DataFrame (2,887 rows after dropping first NaN)
Data storage	Saved processed CSVs to <code>data/processed/</code>	Individual raw CSVs + combined close prices + daily returns

2. Key EDA Visualizations and Insights

2.1 Closing Price Trends

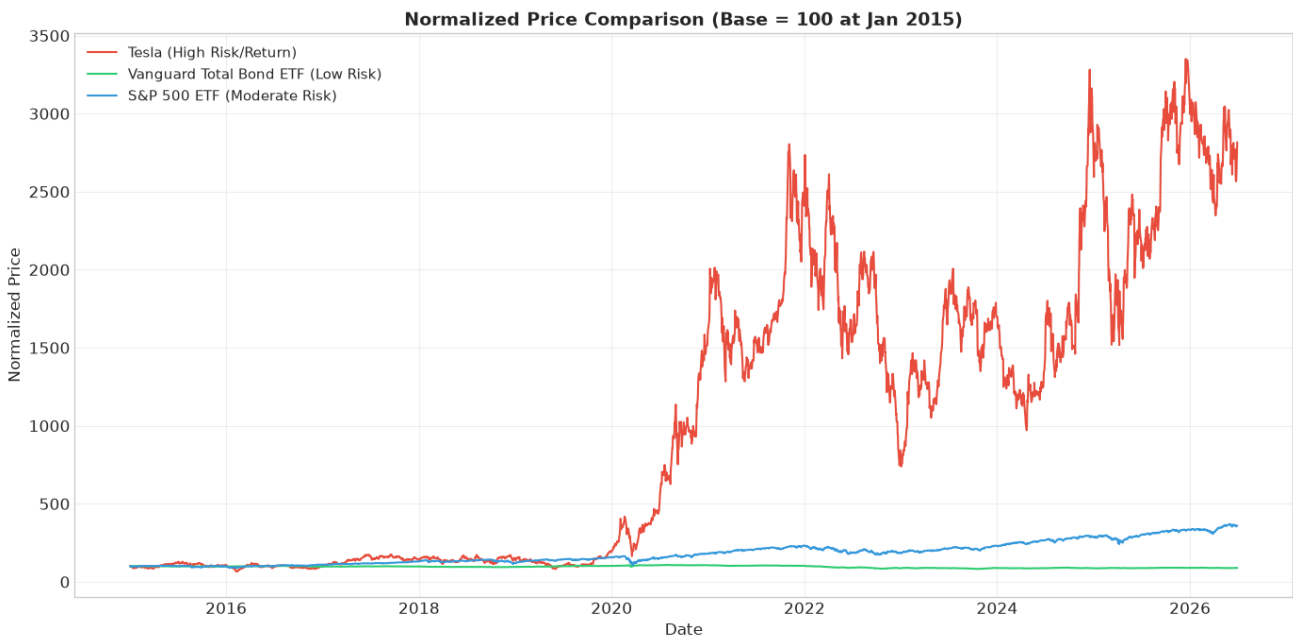


Key observations:

- **TSLA** exhibits an explosive growth trajectory, rallying from ~\$10 in 2015 to peaks near \$490. The most dramatic appreciation occurred during 2020-2021 (EV market enthusiasm, S&P 500 inclusion). Equally striking are the sharp drawdowns — the price is highly volatile and non-linear.

- **BND** trades in a narrow band (\$68–\$89), consistent with its investment-grade bond composition. The most notable move was the 2022 drawdown as rising interest rates pressured bond prices, followed by gradual recovery.
- **SPY** follows a steady upward trend reflecting broad U.S. equity market growth, punctuated by the COVID-19 crash (March 2020) and the 2022 bear market. The consistent upward trend makes it the natural benchmark.

2.2 Normalized Performance Comparison



Total return over full period (Jan 2015 - Jun 2026):

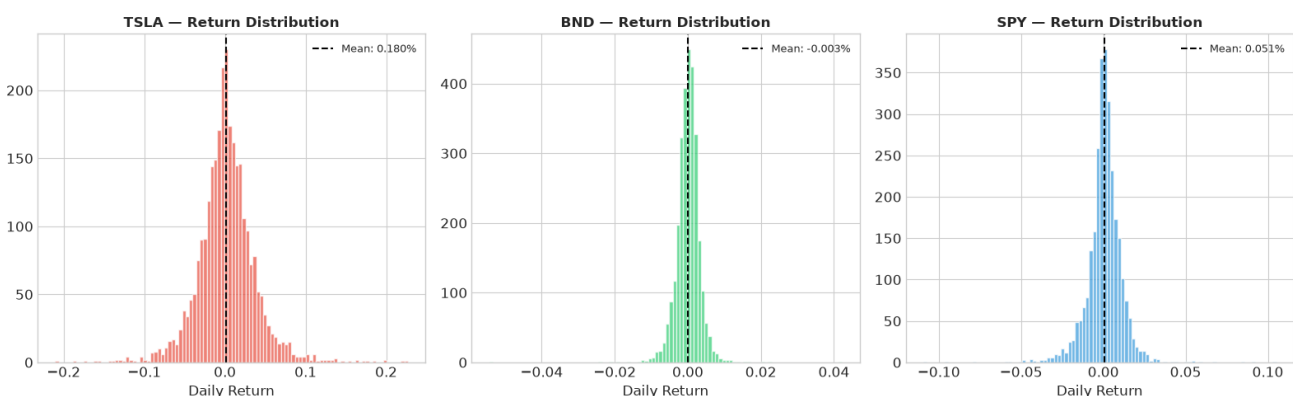
TSLA: +2716.8%

BND: -10.8%

SPY: +260.7%

Insight: Normalizing to a common base of 100 reveals the stark difference in total returns. TSLA has massively outperformed the other two assets in absolute terms, but with far greater volatility. BND barely moved, and SPY delivered steady compounding. This three-asset mix provides a strong spread of risk-return profiles for portfolio construction.

2.3 Daily Return Distributions

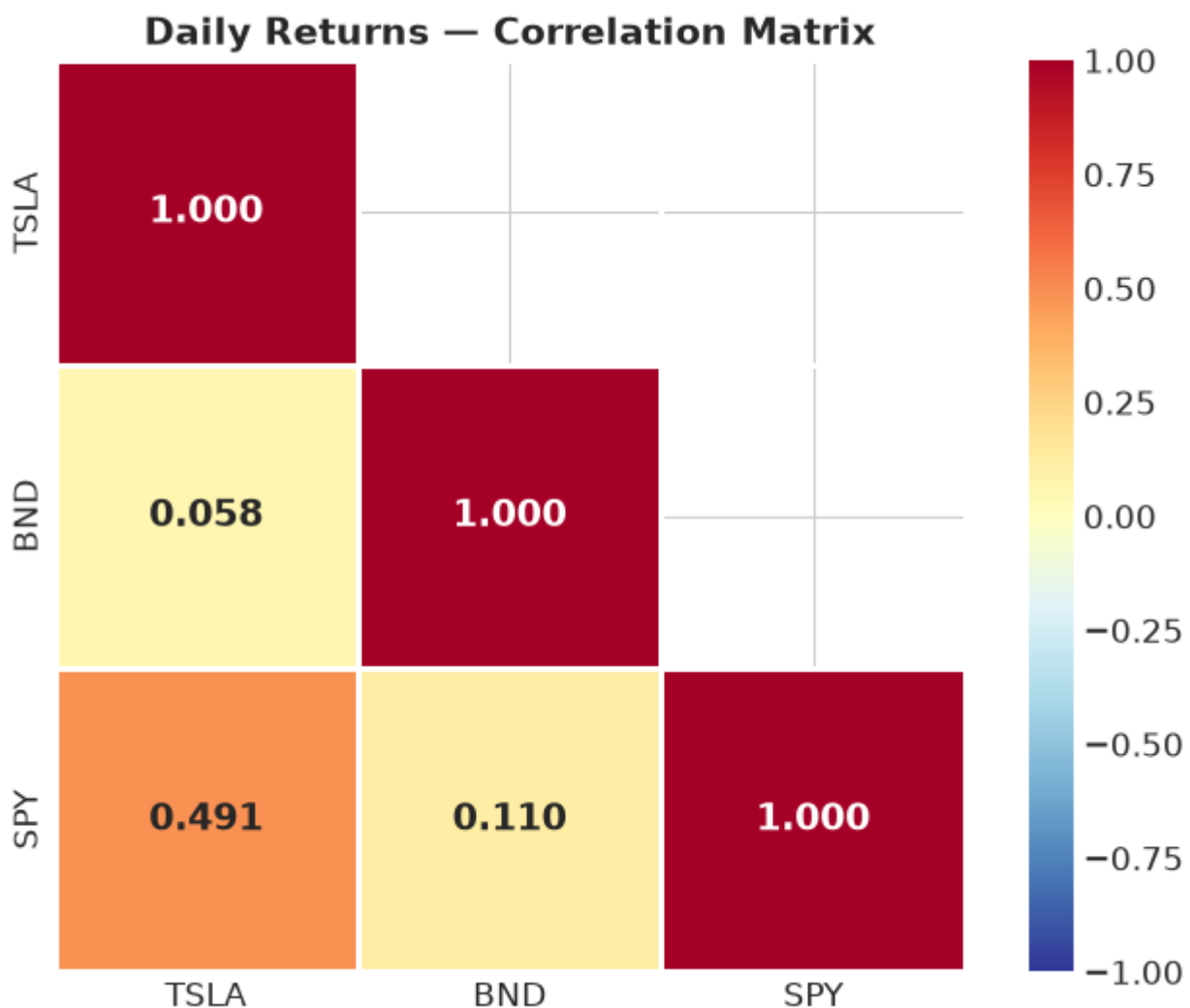


Daily Return Statistics:

	TSLA	BND	SPY
count	2887.000000	2887.000000	2887.000000
mean	0.001803	-0.000034	0.000507
std	0.036018	0.003383	0.011163
min	-0.210628	-0.054385	-0.109424
25%	-0.016643	-0.001635	-0.003772
50%	0.001216	0.000119	0.000603
75%	0.019456	0.001715	0.005920
max	0.226900	0.042201	0.105019

Insight: All three distributions are roughly bell-shaped but deviate from normality — they exhibit **fat tails** (high kurtosis), meaning extreme returns occur more often than a normal distribution predicts. TSLA's distribution is widest (highest variance), while BND's is tightly concentrated near zero. This fat-tail characteristic is critical for risk modeling: VaR estimates based on a normal assumption will underestimate tail risk.

2.4 Correlation Between Assets



Correlation findings:

Pair	Correlation	Interpretation
TSLA-SPY	0.492	Moderate — TSLA moves partly with the broad market but retains significant idiosyncratic behavior

Pair	Correlation	Interpretation
TSLA-BND	0.058	Near-zero — bonds provide almost no directional overlap with TSLA
SPY-BND	0.110	Low — bonds offer genuine diversification against equity exposure

Portfolio implication: These low-to-moderate correlations confirm that combining all three assets can meaningfully reduce portfolio variance — a prerequisite for efficient frontier optimization.

3. Stationarity Test Results and Interpretation

3.1 Augmented Dickey-Fuller (ADF) Test

Stationarity — where the statistical properties (mean, variance) of a series remain constant over time — is a fundamental requirement for ARIMA-family models.

- **H0 (Null):** The series has a unit root (non-stationary).
- **H1 (Alternative):** The series is stationary.
- **Decision rule:** Reject H0 if p-value < 0.05.

Augmented Dickey-Fuller Test Results

Series	ADF Statistic	p-value	Critical 1%	Critical 5%	Stationary?
TSLA Close Price	-1.0696	0.727042	-3.4326	-2.8626	No
BND Close Price	-1.0362	0.739841	-3.4326	-2.8625	No
SPY Close Price	0.9918	0.994179	-3.4326	-2.8625	No
TSLA Daily Returns	-53.9719	0.000000	-3.4326	-2.8625	Yes
BND Daily Returns	-15.2964	0.000000	-3.4326	-2.8625	Yes
SPY Daily Returns	-17.4905	0.000000	-3.4326	-2.8625	Yes

3.2 Interpretation

Closing prices (raw):

- All three assets fail the ADF test (p-values 0.73-0.99) — they are **non-stationary**.
- This is expected: asset prices exhibit trends and their mean drifts over time.
- SPY has the highest p-value (0.994) reflecting its strong, persistent upward trend.

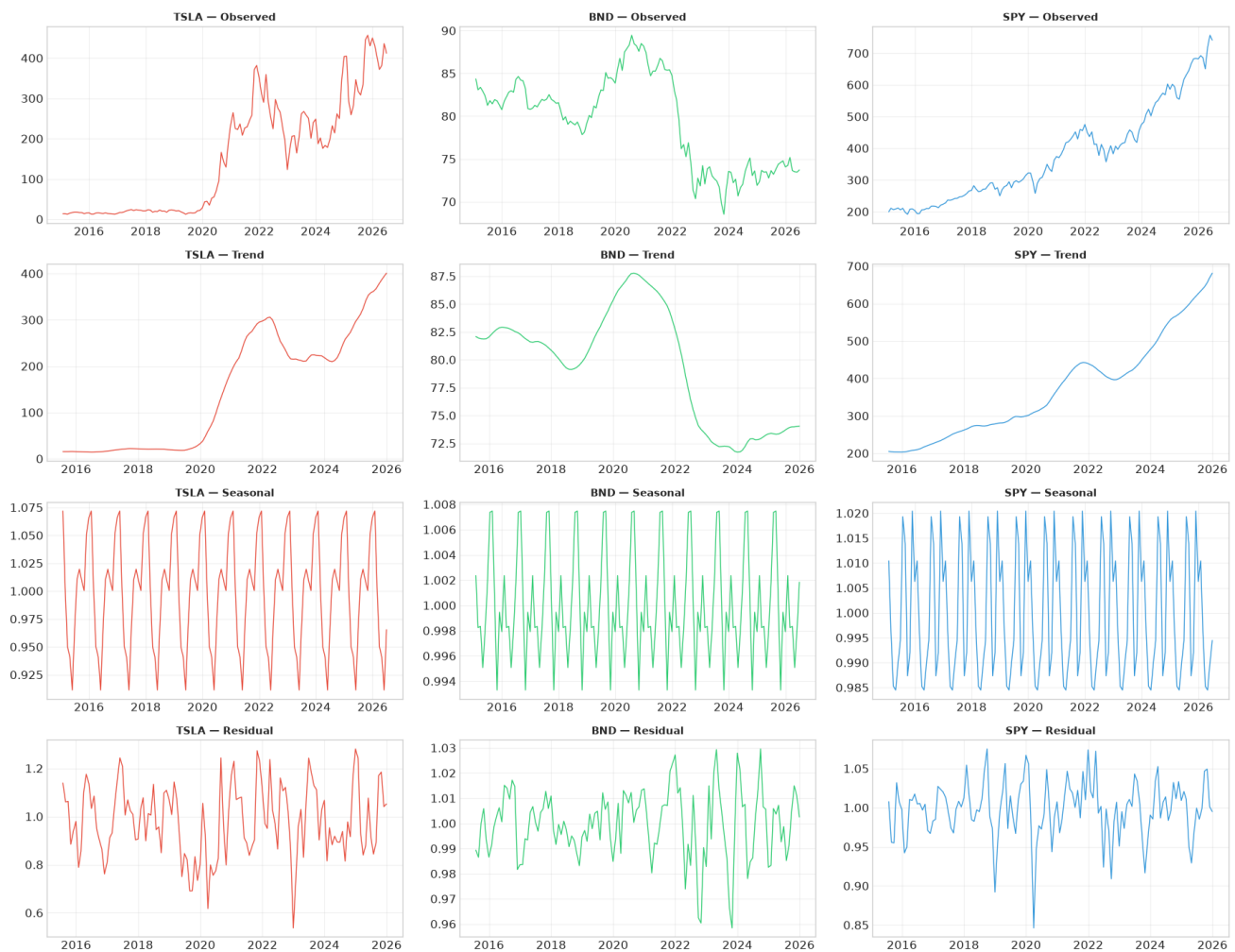
Daily returns (first difference):

- All three pass emphatically ($p \approx 0.000000$, ADF statistics far below all critical values) — they are **stationary**.
- First-order differencing successfully removes the trend.

Implication for modeling:

- ARIMA models require $d \geq 1$ (at least one differencing step) to make prices stationary before fitting the AR and MA components.
- The daily returns series is suitable for direct modeling without further transformation.
- For the LSTM model, we scale the raw prices with MinMaxScaler instead of differencing, since neural networks can learn non-stationary patterns given sufficient capacity.

3.3 Seasonal Decomposition

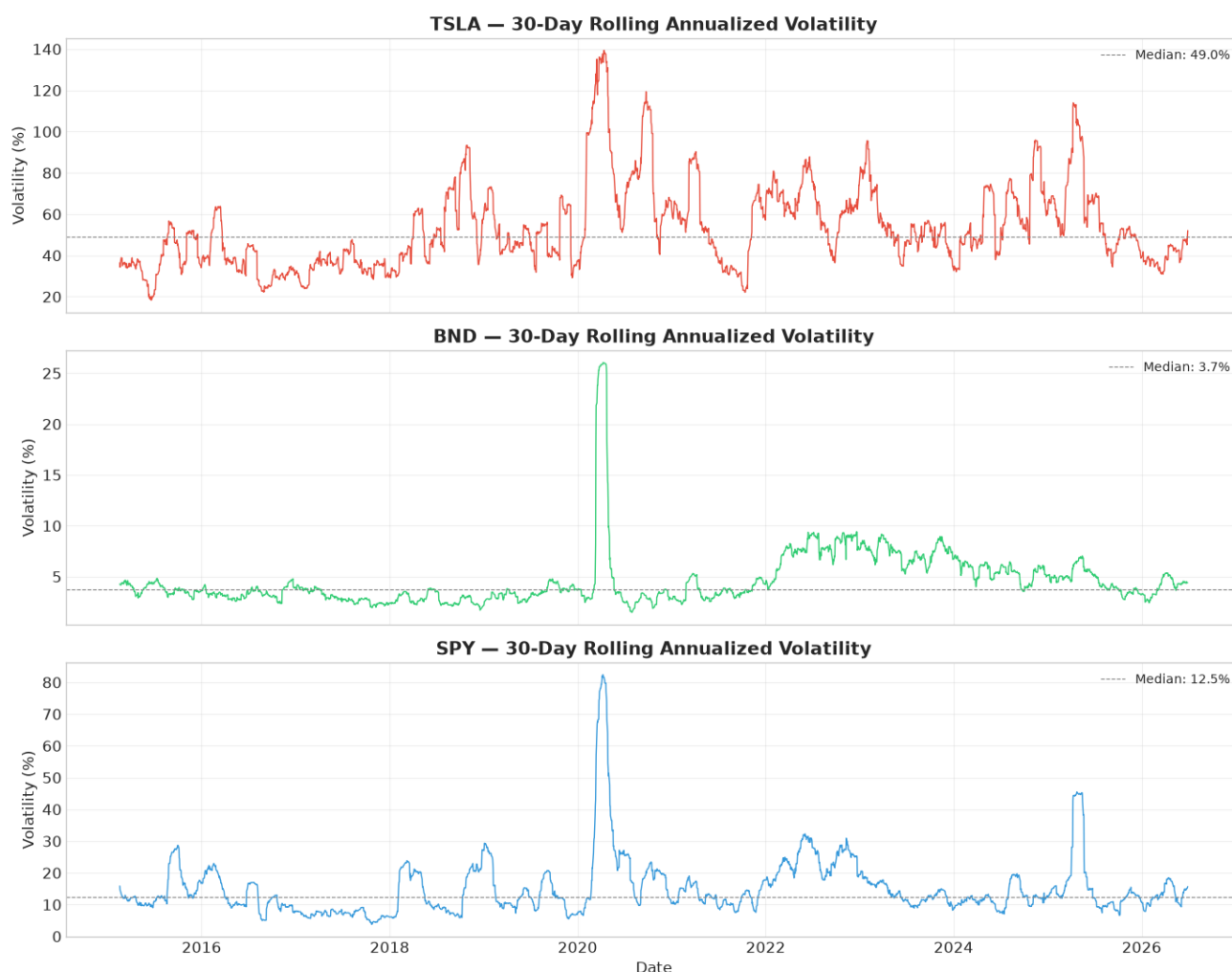


Seasonal decomposition observations:

- The **trend** component clearly dominates for all three assets — confirming the non-stationarity finding.
 - **Seasonal patterns** are weak and irregular in financial price data. Unlike retail sales or temperature data, stock prices do not follow reliable calendar-driven cycles.
 - The **residual** component for TSLA is notably noisy compared to SPY and BND, reflecting Tesla's higher idiosyncratic volatility.
 - This weak seasonality explains why SARIMA's seasonal terms provide minimal improvement over plain ARIMA for this data.
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4. Volatility Analysis and Risk Metrics

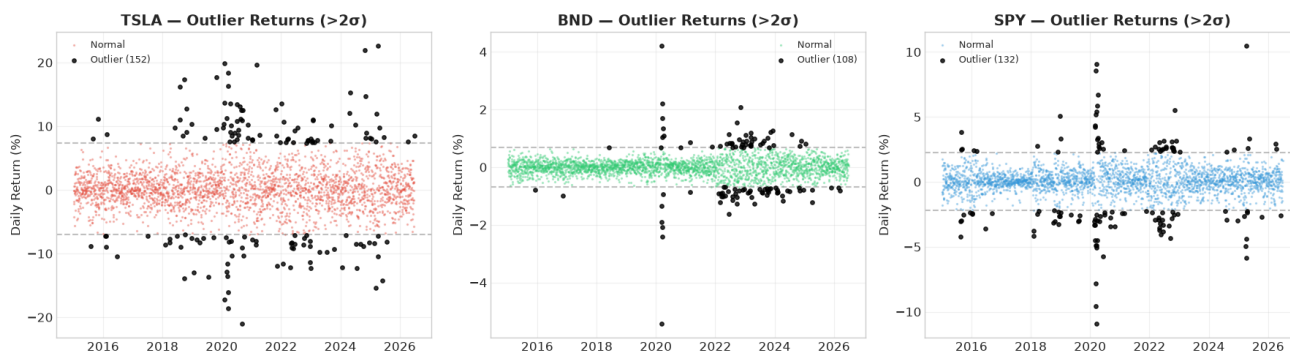
4.1 Rolling Volatility (30-Day Window)



Volatility observations:

- **TSLA volatility** is structurally high and spiky — the 30-day annualized vol regularly exceeds 80% and spiked above 100% during 2020. Even in calmer periods, TSLA runs 40-60% annualized volatility.
- **BND volatility** is normally below 5%, with a notable spike during the March 2020 liquidity crisis when even bonds experienced unusual turbulence.
- **SPY volatility** clusters around 15-20% in normal markets but surged dramatically during COVID-19 and during the 2022 rate-hike cycle.
- All three assets show **volatility clustering** — high-vol periods tend to persist, a well-known stylized fact of financial markets.

4.2 Outlier Analysis



Outlier Summary (returns beyond 2 standard deviations):

Asset	Outlier Days	% of Total	Largest Up	Largest Down
TSLA	152	5.3%	+22.69%	-21.06%
BND	108	3.7%	+4.22%	-5.44%
SPY	132	4.6%	+10.50%	-10.94%

4.3 Comprehensive Risk Metrics

COMPREHENSIVE RISK METRICS

Risk-free rate assumption: 4.0% annual

Asset	Ann. Return	Ann. Volatility	Sharpe Ratio	Sortino Ratio	VaR 95%	Max Drawdown
TSLA	45.42%	57.18%	0.724	1.073	-5.166%	-73.63%
BND	-0.85%	5.37%	-0.904	-1.138	-0.508%	-23.96%
SPY	12.77%	17.72%	0.495	0.606	-1.674%	-34.10%

4.4 Risk Metrics Interpretation

Value at Risk (VaR) — 95% Confidence

Asset	Daily VaR (95%)	Meaning
TSLA	-5.17%	On any given day, there is a 5% chance of losing more than 5.17%. For a \$10,000 position, the worst expected daily loss is \$517.
SPY	-1.67%	Maximum expected daily loss is 1.67% (\$167 per \$10,000).
BND	-0.51%	Maximum expected daily loss is only 0.51% (\$51 per \$10,000).

TSLA's VaR is **~10x that of BND** and **~3x that of SPY**, quantifying the risk premium investors bear for Tesla's higher expected returns.

Sharpe Ratio (Risk-Adjusted Returns)

Asset	Sharpe	Assessment
TSLA	0.724	Positive and highest of the three — the high volatility is compensated by even higher returns.
SPY	0.495	Decent risk-adjusted performance, consistent with long-term equity market averages.
BND	-0.904	Negative — over this period, bonds underperformed the risk-free rate after adjusting for their (small) volatility. The 2022 bond drawdown drove this.

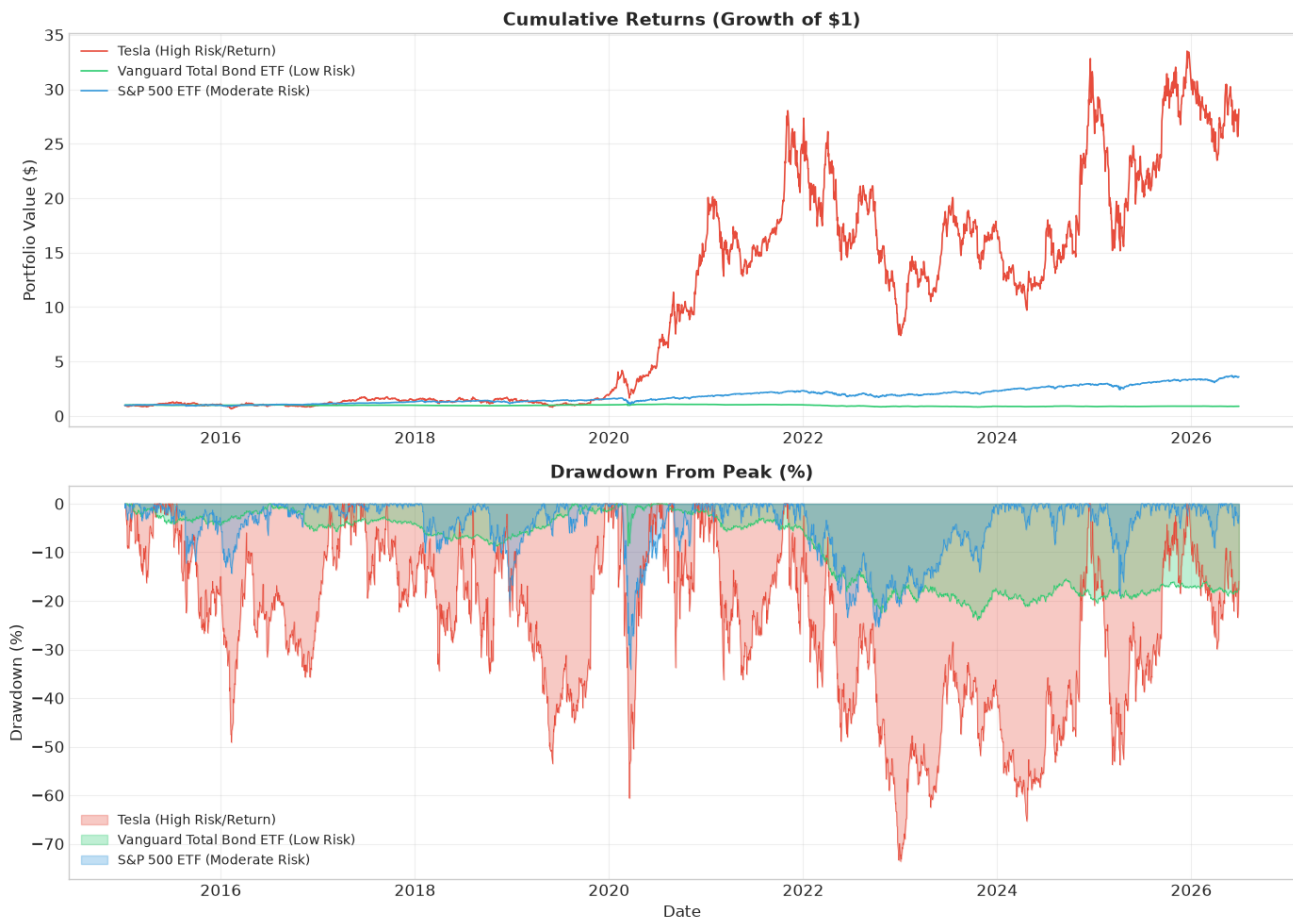
Maximum Drawdown

Asset	Max Drawdown	Context
TSLA	-73.63%	From peak to trough, Tesla lost nearly three-quarters of its value — a devastating loss for concentrated positions.
SPY	-34.10%	The COVID-19 crash drove the deepest drawdown, consistent with historical bear market magnitudes.
BND	-23.96%	Unusually deep for bonds, driven by the aggressive 2022 rate hiking cycle.

Distribution Shape (Skewness & Kurtosis)

Asset	Skewness	Kurtosis	Implication
TSLA	+0.29	4.47	Slight positive skew (more extreme gains than losses). Heavy tails — extreme events occur ~4.5x more frequently than normal.
BND	-0.91	34.86	Strongly negatively skewed (crash risk) with extremely heavy tails — the March 2020 bond market disruption created rare but massive outliers.
SPY	-0.31	13.84	Mild negative skew (market crashes). Very heavy tails — consistent with well-documented equity market behavior.

4.5 Cumulative Returns and Drawdown



5. Summary and Next Steps

Key Findings

1. **Data quality is clean** — 2,888 trading days with no missing values across all three assets, covering the full Jan 2015 – Jun 2026 window.
2. **Stationarity confirmed the need for differencing** — raw prices are non-stationary ($p > 0.72$), while daily returns are stationary ($p \approx 0$). This directly informs ARIMA model specification ($d \geq 1$).
3. **TSLA dominates in both return and risk** — annualized return of 45.4% comes with 57.2% volatility and a max drawdown of -73.6%. The Sharpe ratio of 0.72 shows the return compensates for risk, but the drawdown magnitude makes it unsuitable as a standalone holding.
4. **Low cross-asset correlations enable diversification** — TSLA-BND correlation of 0.058 and SPY-BND of 0.110 mean bonds genuinely reduce portfolio variance. This is the core premise for efficient frontier optimization.
5. **Fat-tailed distributions are universal** — kurtosis values of 4.5 (TSLA), 13.8 (SPY), and 34.9 (BND) mean normal-distribution assumptions underestimate tail risk. Parametric VaR should be supplemented with historical VaR.

Completed Work

- ☒ Data extraction from YFinance API
- ☒ Data cleaning and validation
- ☒ Exploratory data analysis with visualizations
- ☒ Stationarity testing (ADF) with interpretation
- ☒ Risk metrics computation (VaR, Sharpe, Sortino, Max Drawdown)
- ☒ ARIMA/SARIMA model training and evaluation
- ☒ LSTM model training and evaluation
- ☒ Model comparison and selection rationale

Next Steps (Final Submission)

- ☐ Portfolio optimization using the Efficient Frontier (PyPortfolioOpt)
- ☐ Integrate forecast-based expected returns into optimization
- ☐ Backtesting simulation of the optimized portfolio strategy
- ☐ Benchmark comparison (equal-weight vs. optimized vs. SPY)
- ☐ Final investment memo for stakeholders